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Beam indication for inter-cell multiple transmission and reception (multi-TRP, MTRP) operation

Abstract

To support inter-cell multi-TRP operation, disclosed are embodiments that provide a beam indication for the common signals, e.g., signals transmitted and scheduled by some CSS. The disclosed embodiments provide inter-cell multi-TRP with or without serving cell change. In addition, some embodiments provide for switching between inter-cell multi-TRP and intra-cell multi-TRP.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0140776	12/2018	Seo	N/A	H04B 7/0456
2020/0350957	12/2019	Zhou et al.	N/A	N/A
2022/0322310	12/2021	Laddu	N/A	H04W 72/23
2022/0408383	12/2021	Kurita	N/A	H04L 5/005
2022/0417909	12/2021	Zhu	N/A	H04W 16/28
2023/0163914	12/2022	Koskela	370/242	H04W 24/08
2023/0164800	12/2022	Khoshnevisan	370/329	H04L 5/0035
2023/0188293	12/2022	Liu	370/329	H04W 56/0015
2024/0040589	12/2023	Wu	N/A	H04W 72/232
2024/0235783	12/2023	Matsumura	N/A	H04L 5/0094

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113259952	12/2020	CN	N/A

OTHER PUBLICATIONS

Apple Inc. , “On Beam Management Enhancement”, R1-2006499, 3GPP TSG-RAN WG1 Meeting #102-e, e-Meeting, Agenda Item 8.1.1, Aug. 17-20, 2020, 10 pages. cited by applicant

Moderator (Samsung) , “Moderator summary for multi-beam enhancement: proposal categorization”, R1-2006985, 3GPP TSG RAN WG1 #102-e, e-Meeting, Agenda Item 8.1.1, Aug. 17-28, 2020, 27 pages. cited by applicant

OPPO , “Enhancements on beam management for multi-TRP”, R1-2005986, 3GPP TSG RAN WG1 #102-e, e-Meeting, Agenda Item 8.1.2.3, Aug. 17-28, 2020, 3 pages. cited by applicant

PCT/CN2021/120461 , International Search Report and Written Opinion, Apr. 25, 2022, 9 pages. cited by applicant

Mediatek Inc., “Enhancement on multi-beam operation”, R1-2107485, 3GPP TSG RAN WG1 #106-e, e-Meeting, Agenda Item 8.1.1, Aug. 16-27, 2021, 37 pages. cited by applicant

Nokia, Nokia Shanghai Bell, “Enhancements on Multi-beam Operation”, R1-2108052, 3GPP TSG RAN WG1 #106-e, e-Meeting, Agenda Item 8.1.1, Aug. 16-27, 2021, 30 pages. cited by applicant

ZTE, “Enhancements on Multi-beam Operation”, R1-2106541, 3GPP TSG RAN WG1 Meeting #106-e, e-Meeting, Agenda Item 8.1.1, Aug. 16-27, 2021, 24 pages. cited by applicant

Background/Summary

TECHNICAL FIELD

(1) This application relates generally to wireless communication systems, including inter-cell mTRP.

BACKGROUND OF THE DISCLOSURE

(2) Wireless mobile communication technology uses various standards and protocols to transmit data between a base station and a wireless communication device. Wireless communication system standards and protocols can include, for example, 3rd Generation Partnership Project (3GPP) long term evolution (LTE) (e.g., 4G), 3GPP new radio (NR) (e.g., 5G), and IEEE 802.11 standard for wireless local area networks (WLAN) (commonly known to industry groups as Wi-Fi®).

(3) As contemplated by the 3GPP, different wireless communication systems standards and protocols can use various radio access networks (RANs) for communicating between a base station of the RAN (which may also sometimes be referred to generally as a RAN node, a network node, or simply a node) and a wireless communication device known as a user equipment (UE). 3GPP RANs can include, for example, global system for mobile communications (GSM), enhanced data rates for GSM evolution (EDGE) RAN (GERAN), Universal Terrestrial Radio Access Network (UTRAN), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), and/or Next-Generation Radio Access Network (NG-RAN).

(4) Each RAN may use one or more radio access technologies (RATs) to perform communication between the base station and the UE. For example, the GERAN implements GSM and/or EDGE RAT, the UTRAN implements universal mobile telecommunication system (UMTS) RAT or other 3GPP RAT, the E-UTRAN implements LTE RAT (sometimes simply referred to as LTE), and NG-RAN implements NR RAT (sometimes referred to herein as 5G RAT, 5G NR RAT, or simply NR). In certain deployments, the E-UTRAN may also implement NR RAT. In certain deployments, NG-RAN may also implement LTE RAT.

(5) A base station used by a RAN may correspond to that RAN. One example of an E-UTRAN base station is an Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Node B (also commonly denoted as evolved Node B, enhanced Node B, eNodeB, or eNB). One example of an NG-RAN base station is a next generation Node B (also sometimes referred to as a g Node B or gNB).

(6) A RAN provides its communication services with external entities through its connection to a core network (CN). For example, E-UTRAN may utilize an Evolved Packet Core (EPC), while NG-RAN may utilize a 5G Core Network (5GC).

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

(2) FIG. 1 is a block diagram showing an example architecture of a wireless communication system, according to embodiments disclosed herein.

(3) FIG. 2 is a block diagram showing multiple cells for inter-cell mTRP, according to

embodiments disclosed herein.

(4) FIG. 3 is a block diagram showing switching between intra-cell mTRP and intra-cell mTRP, according to embodiments disclosed herein.

(5) FIG. 4 is a flow diagram of a process, in accordance with one embodiment.

(6) FIG. 5 is a table showing a definition of PDCCH-ConfigCommon and PDSCH-ConfigCommon, according to embodiments disclosed herein.

(7) FIG. 6 is a table showing a definition of PUCCH-ConfigCommon, according to embodiments disclosed herein.

(8) FIG. 7 is a table showing a definition of RACH-ConfigCommon, according to embodiments disclosed herein.

(9) FIG. 8 is a flow diagram of a process, in accordance with one embodiment.

(10) FIG. 9 is a block diagram showing a system for performing signaling between a wireless device and a network device, according to embodiments disclosed herein.

DETAILED DESCRIPTION

(11) Various embodiments are described with regard to a UE. However, reference to a UE is merely provided for illustrative purposes. The example embodiments may be utilized with any electronic component that may establish a connection to a network and is configured with the hardware, software, and/or firmware to exchange information and data with the network. Therefore, the UE as described herein is used to represent any appropriate electronic component.

(12) FIG. 1 illustrates an example architecture of a wireless communication system **100**, according to embodiments disclosed herein. The following description is provided for an example wireless communication system **100** that operates in conjunction with the LTE system standards and/or 5G or NR system standards as provided by 3GPP technical specifications.

(13) As shown by FIG. 1, wireless communication system **100** includes UE **102** and UE **104** (although any number of UEs may be used). In this example, UE **102** and UE **104** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device configured for wireless communication.

(14) UE **102** and UE **104** may be configured to communicatively couple with a RAN **106**. In embodiments, RAN **106** may be NG-RAN, E-UTRAN, etc. UE **102** and UE **104** utilize connections (or channels) (shown as connection **108** and connection **110**, respectively) with RAN **106**, each of which comprises a physical communications interface. RAN **106** can include one or more base stations, such as base station **112** and base station **114**, that enable connection **108** and connection **110**.

(15) In this example, connection **108** and connection **110** are air interfaces to enable such communicative coupling, and may be consistent with RAT(s) used by RAN **106**, such as, for example, an LTE and/or NR.

(16) In some embodiments, UE **102** and UE **104** may also directly exchange communication data via a sidelink interface **116**. UE **104** is shown to be configured to access an access point (shown as AP **118**) via connection **120**. By way of example, connection **120** can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein AP **118** may comprise a Wi-Fi® router. In this example, AP **118** may be connected to another network (for example, the Internet) without going through a CN **122**.

(17) In embodiments, UE **102** and UE **104** can be configured to communicate using orthogonal frequency division multiplexing (OFDM) communication signals with each other or with base station **112** and/or base station **114** over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an orthogonal frequency division multiple access (OFDMA) communication technique (e.g., for downlink communications) or a single carrier frequency division multiple access (SC-FDMA) communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the embodiments is not

limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

(18) In some embodiments, all or parts of base station **112** or base station **114** may be implemented as one or more software entities running on server computers as part of a virtual network. In addition, or in other embodiments, base station **112** or base station **114** may be configured to communicate with one another via interface **124**. In embodiments where wireless communication system **100** is an LTE system (e.g., when CN **122** is an EPC), interface **124** may be an X2 interface. The X2 interface may be defined between two or more base stations (e.g., two or more eNBs and the like) that connect to an EPC, and/or between two eNBs connecting to the EPC. In embodiments where wireless communication system **100** is an NR system (e.g., when CN **122** is a 5GC), interface **124** may be an Xn interface. The Xn interface is defined between two or more base stations (e.g., two or more gNBs and the like) that connect to 5GC, between a base station **112** (e.g., a gNB) connecting to 5GC and an eNB, and/or between two eNBs connecting to 5GC (e.g., CN **122**).

(19) RAN **106** is shown to be communicatively coupled to CN **122**. CN **122** may comprise one or more network elements **126**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UE **102** and UE **104**) who are connected to CN **122** via RAN **106**. The components of CN **122** may be implemented in one physical device or separate physical devices including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium).

(20) In embodiments, CN **122** may be an EPC, and RAN **106** may be connected with CN **122** via an S1 interface **128**. In embodiments, S1 interface **128** may be split into two parts, an S1 user plane (S1-U) interface, which carries traffic data between base station **112** or base station **114** and a serving gateway (S-GW), and the S1-MME interface, which is a signaling interface between base station **112** or base station **114** and mobility management entities (MMEs).

(21) In embodiments, CN **122** may be a 5GC, and RAN **106** may be connected with CN **122** via an NG interface **128**. In embodiments, NG interface **128** may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between base station **112** or base station **114** and a user plane function (UPF), and the S1 control plane (NG-C) interface, which is a signaling interface between base station **112** or base station **114** and access and mobility management functions (AMFs).

(22) Generally, an application server **130** may be an element offering applications that use internet protocol (IP) bearer resources with CN **122** (e.g., packet switched data services). Application server **130** can also be configured to support one or more communication services (e.g., VoIP sessions, group communication sessions, etc.) for UE **102** and UE **104** via CN **122**. Application server **130** may communicate with CN **122** through an IP communications interface **132**.

(23) Downlink control information (DCI) in 5G/NR has a purpose that is the same as that of DCI in LTE. That is, DCI is a set of information which schedules downlink data channel (e.g., physical downlink shared channel, PDSCH) or uplink data channel (e.g., physical uplink shared channel, PUSCH).

(24) Physical downlink control channel (PDCCH) search space refers to the area in the downlink resource grid where PDCCH may be carried. A UE performs blind decoding throughout these search space trying to find PDCCH data (i.e., DCI). In order for UE to decode PDCCH (DCI), it determines the exact value for location, structure, scrambling code, and etc. This information is not informed to UE beforehand and in most cases the values change dynamically. The only thing known to UE is the information about a certain range that possibly carries PDCCH (DCI). The UE performs blind decoding in a predefined region. The predefined region in which UE perform the blind decoding is called a search space.

(25) There are two types of search space called a UE-specific search space (USS) and a common search space (CSS). The USS is dedicated for each specific UE and informed to UE via RRC signaling message. It means UE needs to complete RRC establishment to get the information about

the USS. The common search space (CSS) is the search space that every UE or group of UEs (group casts) needs to search for the signals for every UE (e.g., PDCCH for SIB) or signaling message that is applied to every UE before dedicated channel is established for a specific UE (e.g., PDCCH used during RACH process). For example, a UE needs to detect PDCCH for SIB1 reception or various DCI (PDCCH) during RACH process (e.g., DCI for Msg2/Msg4 reception).

(26) In Release 16 of the 3GPP standards (Rel-16), intra-cell multi-TRP operation has been supported with two modes: single- and multi-DCI modes. Single-DCI mode is when a gNB can schedule PDSCH from two TRPs by a single DCI. Multi-DCI mode is when PDSCH from two TRPs are scheduled by two DCIs. The two DCIs are transmitted in control resource sets (CORESETs) with different CORESETPoolIndex.

(27) The gNB can provide beam indication, e.g., transmission configuration indicator (TCI) indication, for CORESET and PDSCH based on channel state information reference signal (CSI-RS). A synchronization signal block (SSB) can be indicated as the quasi-co-location (QCL) source for the CSI-RS in its TCI source.

(28) SS/CORESET 0 is a special SS/CORESET, in which each instance is associated with an SSB. PDCCH beam, time/frequency location are determined by the associated SSB.

(29) A UE need not monitor all instances for SS/CORESET 0. Instead, a UE would monitor the SS/CORESET 0 instance associated with one SSB from the most recent of the following: SSB associated with RACH procedure or SSB QCLed with the CSI-RS in the TCI state for the CORESET 0.

(30) In terms of search space configuration, there is a total of six types of search spaces. There are five types of CSS and one type of USS.

(31) CSS includes the following types described under section 10.1 of 3GPP 38.213: Type0-PDCCH CSS set; Type0A-PDCCH CSS set; Type1-PDCCH CSS set; Type2-PDCCH CSS set; and Type3-PDCCH CSS set. For CSS, some common signals are transmitted or scheduled by some CSS. A common signals is one that is received by multiple UEs, for example, PDCCH transmitted in CSS sets not associated with cell radio network temporary identifier (RNTI) (C-RNTI), modulation coding scheme (MCS)C-RNTI (MCS-C-RNT), as well as corresponding PDSCH scheduled by the PDCCH.

(32) Type0-PDCCH CSS set is configured by `pdccch-ConfigSIB1` in MIB or by `searchSpaceSIB1` in `PDCCH-ConfigCommon` or by `searchSpaceZero` in `PDCCH-ConfigCommon` for a DCI format with cyclic redundancy check (CRC) scrambled by a system information (SI) radio network temporary identifier (RNTI) (SI-RNTI) on the primary cell of the master cell group (MCG).

(33) Type0A-PDCCH CSS set is configured by `searchSpaceOtherSystemInformation` in `PDCCH-ConfigCommon` for a DCI format with CRC scrambled by a SI-RNTI on the primary cell of the MCG.

(34) Type1-PDCCH CSS set is configured by `ra-SearchSpace` in `PDCCH-ConfigCommon` for a DCI format with CRC scrambled by a random access (RA) RNTI (RA-RNTI), a `MsgB-RNTI`, or a temporary cell (TC) RNTI (TC-RNTI) on the primary cell.

(35) Type2-PDCCH CSS set is configured by `pagingSearchSpace` in `PDCCH-ConfigCommon` for a DCI format with CRC scrambled by a paging RNTI (P-RNTI) on the primary cell of the MCG.

(36) Type3-PDCCH CSS set is configured by `SearchSpace` in `PDCCH-Config` with `searchSpaceType=common` for DCI formats with CRC scrambled by interruption RNTI (INT-RNTI), slot format indication (SFI) RNTI (SFI-RNTI), transmit power control (TPC) PUSCH RNTI (TPC-PUSCH-RNTI), TPC-PUCCH RNTI, TPC sounding reference symbols (SRS) RNTI (TPC-SRS-RNTT), or cancellation indication (CI) RNTI (CI-RNTI) and, only for the primary cell, C-RNTI, MCS-C-RNTI, configured scheduling (CS) RNTI(s) (CS-RNTI), or power saving (PS) RNTI (PS-RNTI).

(37) USS is for some dedicated transmission. For instance, a dedicated signal is a PDCCH associated with a C-RNTI/MCS-C-RNTI or a PDSCH scheduled by DCI with C-RNTI/MCS-C-

RNTI.

(38) In Release 17 of the 3GPP standards (Rel-16), the multi-DCI operation is to be enhanced to support inter-cell operation, as shown in FIG. 2. FIG. 2 is a diagram of inter-cell multi-TRP **200** in which a UE **202** is connected to a first gNB **204** acting a first cell **206** and a second gNB **208** acting as a second cell **210**. A third gNB **212** providing a third cell **214** is also available.

(39) In inter-cell multi-TRP **200**, a PDSCH can be transmitted from different cells with different cell ID. SSB pattern in different cells could be the same or different. SSBs from different cells may be multiplexed in the same symbols. The PDSCH from different cells may be overlapped in time/frequency domain. Accordingly, an issue in supporting inter-cell multi-TRP operation is how to provide the beam indication for the common signals, e.g., signals transmitted and scheduled by the CSS. In particular, the issue arises in the following two cases. A first case is inter-cell multi-TRP without a serving cell change. A second case is inter-cell multi-TRP with a serving cell change. In addition, a related issue is how to support switching between inter-cell multi-TRP and intra-cell multi-TRP, which should also consider the two cases.

(40) In the first case (without serving cell change), the common signal is QCLed with one of the SSB associated with the serving cell physical cell ID (PCI). A serving cell is the cell where UE decoded the master information block (MIB). In some embodiments, there are two options described as follows.

(41) A first option is when the CORESETs are divided into two groups. A first group (group 1) includes CORESETs associated with some or all just CSS, or at least associated with some or all CSS. It may include Type 0/0A/1/2 CSS or all CSS. The TCI state associated with serving cell PCI can be used for beam indication. As an extension, the QCL assumption for these CORESETs can also be updated by SSB associated with some random access procedure, e.g., in response to the UE sending to the gNB a physical random access channel (PRACH) associated with SSB in contention-based random access (CBRA). A second group (group 2) includes CORESETs associated with a USS and optionally Type 3 CSS. For this second group, the TCI state associated with a PCI from the assistant cell can be used for beam indication because dedicated signal can still be transmitted from assistant cells. An assistant cell is the cell that can communicate with the UE in inter-cell multi-TRP mode.

(42) The PDSCH scheduled by CORESETs in group 1 can share the same beam as the CORESETs. When the UE decodes the PDSCH corresponding to the CORESETs in group 1, it can prioritize the beam to receive such PDSCH, regardless of whether there is another PDSCH scheduled by CORESETs in group 2 or not. In one example, group 1 may only include CORESET 0.

(43) A second option is when the TCI indication is not applied for some or all CSS. In one example, TCI indication is not applied for Type 0/0A/1/2 CSS. In another example, the TCI indication is not applied for all CSS. The QCL assumption for the precluded CSS can be updated by SSB associated with some random access procedure, e.g., contention-based random access (CBRA). The PDSCH scheduled by the precluded CSS can share the same beam as the precluded CSS, meaning both common PDCCH and PDSCH are based on the same beam from serving cell. When the UE decodes the PDSCH corresponding to the CSS, it can prioritize the beam to receive such PDSCH, regardless of whether there is another PDSCH scheduled by another search space or not.

(44) With regard to switching between inter-cell mTRP and intra-cell mTRP, all the CORESETPoolIndex should only be associated with the serving cell PCI when UE is switched to intra-cell mTRP. When UE is switched to inter-cell mTRP, at least one CORESETPoolIndex should be associated with the serving cell PCI and different CORESETPoolIndex should be associated with different PCIs. Association between CORESETPoolIndex and PCI indicates that signals scheduled or transmitted from CORESET with a CORESETPoolIndex should be QCLed directly or indirectly with SSBs with the PCI.

(45) FIG. 3 shows an example of mTRP **300** for dynamic cell switching and QCL configuration

without serving cell change. An upper portion of FIG. 3 shows inter-cell mTRP **302**. A lower portion of FIG. 3 shows intra-cell mTRP **304**.

(46) Inter-cell mTRP **302** includes a first cell **306**, a second cell **308**, and a UE **310**. First cell **306** is assumed to be a serving cell for UE **310**. Common signal **312** comes from CORESETPoolIndex zero, which is associated with first cell **306**. First cell **306** provides a dedicated signal **314**. Second cell **308** provides dedicated signal **316**.

(47) When UE **310** switches to intra-cell mTRP **304**, there are two TRPs in a first cell. Specifically, intra-cell mTRP **304** includes a first TRP **318**, a second TRP **320**, and UE **310** in a cell.

Accordingly, a common signal **322** can come from any CORESETPoolIndex (e.g., zero or one), associated with the same PCI. A dedicated signal **324** and a dedicated signal **326** can still come from both first TRP **318** and second TRP **320**.

(48) FIG. 4 shows a process **400**, performed by a gNB defining a serving cell, of providing to a UE a beam indication for common signals transmitted and scheduled by a CSS for inter-cell multi-TRP operation without a change of the serving cell. In block **402**, process **400** quasi-co-locates a common signal with an SSB associated with a PCI of the serving cell. In block **404**, process **400** dividing CORESETs into a first group and a second group, the first group being associated with at least some of the CSS and a TCI state associated with the PCI of the serving cell is used for beam indication, the second group being associated with a USS and a TCI state associated with a PCI of any cell (i.e., serving cell or assistant cells(s)) is used for beam indication. In block **406**, process **400** schedules a PDSCH by a CORESET in the first group. Process **400** may also include the first group comprised of type 0, type 0A, type 1, type 2 CSS, or all types of CSS. And in some embodiments, the second group includes a type 3 CSS. In another embodiment, the first group includes only CORESET 0. Process **400** may also include, in response to the UE sending to the gNB a PRACH associated with the SSB, updating a quasi-co-location for the first group. Process **400** may also include the PDSCH being scheduled by a PDCCH in CORESETs in the first or second groups and having a same beam with CORESETs in the first or second groups. Process **400** may also include, in response to the UE switching from intra-cell multi-TRP, associating a CORESETPoolIndex with the PCI of the serving cell.

(49) When serving cell change is allowed, the common signal is QCLed with one of the SSB associated with PCI other than serving cell PCI. There are three options for this embodiment.

(50) A first option is that some of or all the parameters present in common configuration under section 6.3.2 of 3GPP 38.331 for the associated cell, e.g., PDCCH-ConfigCommon **502** (FIG. 5), PDSCH-ConfigCommon **504** (FIG. 5), PUSCH-ConfigCommon **602** (FIG. 6), PUCCH-ConfigCommon **604** (FIG. 6), and RACH-ConfigCommon **702** (FIG. 7) can be provided by RRC signaling. When the common signals are associated with the assistant cell PCI, a UE should use the corresponding configuration to receive or transmit the common signal. As an extension, the common signals may be associated with one cell at a time. The options described above (first option is when the CORESETs are divided into two groups; a second option is when the TCI indication is not applied for some or all CSS) may be reused for beam indication, where the indicated beam for the common signal can be associated with one cell.

(51) A second option is when UE assumes the common configuration for assistant cell and serving cell are the same.

(52) A third option is a mix of the first and second options. For instance, the gNB can optionally provide configuration based on the first option for a cell. If the configuration is not provided, UE assumes the configuration for the cell is the same as serving cell.

(53) If a UE communicates with a single cell, e.g., the UE receives or transmits common signal to one cell, and has a serving cell change, then radio resource management (RRM) should consider this cell as the new serving cell. All the measurement events, e.g., event A1-A6 of section 5.5.4 in 3GPP 38.331 should consider the new serving cell measurement results.

(54) If a UE communicates with more than one cells, e.g., UE receive or transmit common signal to

more than one cells, and has a serving cell change, then there are three options for RRM.

(55) A first option is that the RRM considers one cell as serving cell. The cell may be configured by higher layer signaling, e.g., RRC or MAC CE, or predefined. An example of a predefined serving cell is a cell with lowest PCI or cell associated with CORESET with lowest ID or cell associated with CORESETPoolIndex equal to zero.

(56) A second option is that the RRM considers all the cells as serving cells. The event A1-A6 considers the minimal/maximum/average measurement results from all the cells.

(57) A third option is that the RRM procedure can be extended as a per CORESETPoolIndex procedure. The event A1-A6 can be applied per CORESETPoolIndex to assist gNB to make the decision for cell change for a CORESETPoolIndex.

(58) With serving cell change, a gNB may configure the C-RNTI and/or MCS-C-RNTI for the new cell by higher layer signaling, e.g., RRC or MAC CE. If a UE communicates with a single cell, e.g., UE only receive or transmit common signal to one cell, UE should apply the C-RNTI/MCS-C-RNTI associated with the cell for communication. If a UE communicates with more than one cells, e.g., UE receive or transmit common signal to more than one cells, there are two options for C-RNTI/MCS-C-RNTI.

(59) A first option is that UE considers one cell as serving cell. The cell may be configured by higher layer signaling, e.g., RRC or MAC CE, or predefined (cell with lowest PCI or cell associated with CORESET with lowest ID or cell associated with CORESETPoolIndex equal to zero). UE should apply the C-RNTI/MCS-C-RNTI associated with the cell for communication.

(60) A second option is that a UE considers all the cells as serving cells. For signals associated with each cell, UE applies corresponding C-RNTI/MCS-C-RNTI.

(61) In another embodiment, C-RNTI/MCS-C-RNTI can be configured per CORESETPoolIndex.

(62) FIG. 8 shows a processes **800** performed by a UE, of obtaining a beam indication for common signals transmitted and scheduled by CSS for inter-cell multi-TRP operation with a change of a serving cell. In block **802**, process **800** receives from the serving cell RRC signaling of configuration parameters for a common signal of an assistant cell. In block **804**, process **800** in response to a common signal quasi-co-located with an SSB associated with a PCI of the assistant cell, receives the common signal for the assistant cell, in which a beam indication is provided through a TCI state associated with the PCI or is updated by the SSB in response to a random access procedure. Process **800** may also include performing radio resource management measurement events based on a new cell corresponding to the PCI. Process **800** may also include performing radio resource management measurement events based on a cell determined by higher layer signaling. Process **800** may also include performing radio resource management measurement events based on a cell determined by a predefined value of PCI or CORESET ID. Process **800** may also include performing radio resource management measurement events based a function of measurement results from all cells acting as serving cells. Process **800** may also include performing radio resource management measurement events based on per CORESETPoolIndex. Process **800** may also include applying a C-RNTI or MCS-C-RNTI to a new cell corresponding to the assistant cell, in which the C-RNTI or MCS-C-RNTI is optionally determined by higher layer signaling, or is a predefined value of PCI or CORESET ID.

(63) FIG. 9 illustrates a system **900** for performing signaling **902** between a wireless device **904** and a network device **906**, according to embodiments disclosed herein. System **900** may be a portion of a wireless communications system as herein described. Wireless device **904** may be, for example, a UE of a wireless communication system. Network device **906** may be, for example, a base station (e.g., an eNB or a gNB) of a wireless communication system.

(64) Wireless device **904** may include one or more processor(s) **908**. Processor(s) **908** may execute instructions such that various operations of wireless device **904** are performed, as described herein. Processor(s) **908** may include one or more baseband processors implemented using, for example, a central processing unit (CPU), a digital signal processor (DSP), an application specific integrated

circuit (ASIC), a controller, a field programmable gate array (FPGA) device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein.

(65) Wireless device **904** may include a memory **910**. Memory **910** may be a non-transitory computer-readable storage medium that stores instructions **912** (which may include, for example, the instructions being executed by processor(s) **908**). Instructions **912** may also be referred to as program code or a computer program. Memory **910** may also store data used by, and results computed by, processor(s) **908**.

(66) Wireless device **904** may include one or more transceiver(s) **914** that may include radio frequency (RF) transmitter and/or receiver circuitry that use antenna(s) **916** of wireless device **904** to facilitate signaling (e.g., signaling **902**) to and/or from wireless device **904** with other devices (e.g., network device **906**) according to corresponding RATs.

(67) Wireless device **904** may include one or more antenna(s) **916** (e.g., one, two, four, or more). For embodiments with multiple antenna(s) **916**, wireless device **904** may leverage the spatial diversity of such multiple antenna(s) **916** to send and/or receive multiple different data streams on the same time and frequency resources. This behavior may be referred to as, for example, multiple input multiple output (MIMO) behavior (referring to the multiple antennas used at each of a transmitting device and a receiving device that enable this aspect). MIMO transmissions by wireless device **904** may be accomplished according to precoding (or digital beamforming) that is applied at wireless device **904** that multiplexes the data streams across antenna(s) **916** according to known or assumed channel characteristics such that each data stream is received with an appropriate signal strength relative to other streams and at a desired location in the spatial domain (e.g., the location of a receiver associated with that data stream). Certain embodiments may use single user MIMO (SU-MIMO) methods (where the data streams are all directed to a single receiver) and/or multi user MIMO (MU-MIMO) methods (where individual data streams may be directed to individual (different) receivers in different locations in the spatial domain).

(68) In certain embodiments having multiple antennas, wireless device **904** may implement analog beamforming techniques, whereby phases of the signals sent by antenna(s) **916** are relatively adjusted such that the (joint) transmission of antenna(s) **916** can be directed (this is sometimes referred to as beam steering).

(69) Wireless device **904** may include one or more interface(s) **918**. Interface(s) **918** may be used to provide input to or output from wireless device **904**. For example, a wireless device **904** that is a UE may include interface(s) **918** such as microphones, speakers, a touchscreen, buttons, and the like in order to allow for input and/or output to the UE by a user of the UE. Other interfaces of such a UE may be made up of made up of transmitters, receivers, and other circuitry (e.g., other than transceiver(s) **914**/antenna(s) **916** already described) that allow for communication between the UE and other devices and may operate according to known protocols (e.g., Wi-Fi®, Bluetooth®, and the like).

(70) Wireless device **904** may include a beam indication module **920**. Beam indication module **920** may be implemented via hardware, software, or combinations thereof. For example, beam indication module **920** may be implemented as a processor, circuit, and/or instructions **912** stored in memory **910** and executed by processor(s) **908**. In some examples, beam indication module **920** may be integrated within processor(s) **908** and/or transceiver(s) **914**. For example, beam indication module **920** may be implemented by a combination of software components (e.g., executed by a DSP or a general processor) and hardware components (e.g., logic gates and circuitry) within processor(s) **908** or transceiver(s) **914**.

(71) Beam indication module **920** may be used for various aspects of the present disclosure. For example, beam indication module **920** is configured to perform process **800** (FIG. 8).

(72) Network device **906** may include one or more processor(s) **922**. Processor(s) **922** may execute instructions such that various operations of network device **906** are performed, as described herein.

Processor(s) **908** may include one or more baseband processors implemented using, for example, a CPU, a DSP, an ASIC, a controller, an FPGA device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein.

(73) Network device **906** may include a memory **924**. Memory **924** may be a non-transitory computer-readable storage medium that stores instructions **926** (which may include, for example, the instructions being executed by processor(s) **922**). Instructions **926** may also be referred to as program code or a computer program. Memory **924** may also store data used by, and results computed by, processor(s) **922**.

(74) Network device **906** may include one or more transceiver(s) **928** that may include RF transmitter and/or receiver circuitry that use antenna(s) **930** of network device **906** to facilitate signaling (e.g., signaling **902**) to and/or from network device **906** with other devices (e.g., wireless device **904**) according to corresponding RATs.

(75) Network device **906** may include one or more antenna(s) **930** (e.g., one, two, four, or more). In embodiments having multiple antenna(s) **930**, network device **906** may perform MIMO, digital beamforming, analog beamforming, beam steering, etc., as has been described.

(76) Network device **906** may include one or more interface(s) **932**. Interface(s) **932** may be used to provide input to or output from network device **906**. For example, a network device **906** that is a base station may include interface(s) **932** made up of transmitters, receivers, and other circuitry (e.g., other than transceiver(s) **928**/antenna(s) **930** already described) that enables the base station to communicate with other equipment in a core network, and/or that enables the base station to communicate with external networks, computers, databases, and the like for purposes of operations, administration, and maintenance of the base station or other equipment operably connected thereto.

(77) Network device **906** may include a beam indication module **934**. Beam indication module **934** may be implemented via hardware, software, or combinations thereof. For example, beam indication module **934** may be implemented as a processor, circuit, and/or instructions **926** stored in memory **924** and executed by processor(s) **922**. In some examples, beam indication module **934** may be integrated within processor(s) **922** and/or transceiver(s) **928**. For example, beam indication module **934** may be implemented by a combination of software components (e.g., executed by a DSP or a general processor) and hardware components (e.g., logic gates and circuitry) within processor(s) **922** or transceiver(s) **928**.

(78) Beam indication module **934** may be used for various aspects of the present disclosure. For example, beam indication module **934** is configured to perform process **400** (FIG. 4).

(79) For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth herein. For example, a baseband processor as described herein in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth herein. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth herein.

(80) Any of the above described embodiments may be combined with any other embodiment (or combination of embodiments), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

(81) Embodiments and implementations of the systems and methods described herein may include various operations, which may be embodied in machine-executable instructions to be executed by a computer system. A computer system may include one or more general-purpose or special-purpose computers (or other electronic devices). The computer system may include hardware components

that include specific logic for performing the operations or may include a combination of hardware, software, and/or firmware.

(82) It should be recognized that the systems described herein include descriptions of specific embodiments. These embodiments can be combined into single systems, partially combined into other systems, split into multiple systems or divided or combined in other ways. In addition, it is contemplated that parameters, attributes, aspects, etc. of one embodiment can be used in another embodiment. The parameters, attributes, aspects, etc. are merely described in one or more embodiments for clarity, and it is recognized that the parameters, attributes, aspects, etc. can be combined with or substituted for parameters, attributes, aspects, etc. of another embodiment unless specifically disclaimed herein.

(83) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

(84) Although the foregoing has been described in some detail for purposes of clarity, it will be apparent that certain changes and modifications may be made without departing from the principles thereof. It should be noted that there are many alternative ways of implementing both the processes and apparatuses described herein. Accordingly, the present embodiments are to be considered illustrative and not restrictive, and the description is not to be limited to the details given herein, but may be modified within the scope and equivalents of the appended claims.

Claims

1. A method, performed by a base station defining a serving cell, of providing to a user equipment (UE) a beam indication for common signals transmitted and scheduled by common search space (CSS) for inter-cell multiple transmission and reception (multi-TRP) operation without a change of the serving cell, the method comprising: quasi-co-locating a common signal with a system synchronization block (SSB) associated with a physical cell ID (PCI) of the serving cell; dividing control resource sets (CORESETs) into a first group and a second group, the first group being associated with at least some of the CSS and a transmission configuration indicator (TCI) state associated with the PCI of the serving cell is used for beam indication, the second group being associated with a UE specific search space (USS) and a TCI state associated with a PCI of the serving cell or an assistant cell is used for beam indication, in which the first group includes type 0, type 0A, type 1, or type 2 CSS, and wherein a TCI indication is not applied for the type 0, type 0A, type 1, or type 2 CSS; and scheduling a physical downlink shared channel (PDSCH) by a CORESET in the first group.
2. The method of claim 1, in which the first group includes all types of CSS.
3. The method of claim 1 further comprising, in response to the UE sending to the base station a physical random access channel (PRACH) associated with the SSB, updating a quasi-co-location for the first group.
4. The method of claim 1, in which the second group includes a type 3 CSS.
5. The method of claim 1, in which the first group includes only CORESET 0.
6. The method of claim 1, in which the PDSCH is scheduled by a physical downlink control channel (PDCCH) in CORESETs in the first or second groups and shares a same beam with CORESETs in the first or second groups.
7. The method of claim 1, further comprising, in response to the UE switching from intra-cell multi-TRP to inter-cell multi-TRP, associating a CORESETPoolIndex with the PCI of the serving cell.

8. A non-transitory computer-readable storage medium of a base station for configuring the base station to provide to a user equipment (UE) a beam indication for common signals transmitted and scheduled by common search space (CSS) for inter-cell multiple transmission and reception (multi-TRP) operation without a change of a serving cell that is defined by the base station, the computer-readable storage medium including instructions that when executed by a processor of the base station, cause the base station to: quasi-co-locate a common signal with a system synchronization block (SSB) associated with a physical cell ID (PCI) of the serving cell; divide control resource sets (CORESETs) into a first group and a second group, the first group being associated with at least some of the CSS and a transmission configuration indicator (TCI) state associated with the PCI of the serving cell is used for beam indication, the second group being associated with a UE specific search space (USS) and a TCI state associated with a PCI of the serving cell or an assistant cell is used for beam indication, in which the first group includes type 0, type 0A, type 1, or type 2 CSS, and wherein a TCI indication is not applied for the type 0, type 0A, type 1, or type 2 CSS; and schedule a physical downlink shared channel (PDSCH) by a CORESET in the first group.
9. The computer-readable storage medium of claim 8, in which the first group includes all types of CSS.
10. The computer-readable storage medium of claim 8, in which the instructions further cause the base station to, in response to the UE send to the base station a physical random access channel (PRACH) associated with the SSB, update a quasi-co-location for the first group.
11. The computer-readable storage medium of claim 8, in which the second group includes a type 3 CSS.
12. The computer-readable storage medium of claim 8, in which the first group includes only CORESET 0.
13. The computer-readable storage medium of claim 8, in which the PDSCH is scheduled by a physical downlink control channel (PDCCH) in CORESETs in the first or second groups and shares a same beam with CORESETs in the first or second groups.
14. The computer-readable storage medium of claim 8, in which the instructions further cause the base station to, in response to the UE switch from intra-cell multi-TRP to inter-cell multi-TRP, associate a CORESETPoolIndex with the PCI of the serving cell.
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